

CPE477/ECG677 – Embedded Security & Machine Learning

Design Assignment 1

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Github Repository link (root): <https://github.com/autonomize009/autonomize009-Assign1>

Youtube Playlist link (root):

<https://www.youtube.com/playlist?list=PLVIuzDVvAe9iHvuCgkgMcRMypPbO1yoZG>

Follow the submission guideline to be awarded points for this Assignment.

Submit the following for all Assignments:

1. In the document, for each task submit the modified or included code (from the base code) with highlights and justifications of the modifications. Also include the comments. If no base code is provided, submit the base code for the first task only.
2. Create a private Github repository with a random name (no CPE477/677, Lastname, Firstname). Place all assignments under the root folder, sub-folder named Assignmentn, with one document and one video link file for each lab, place modified c files named as main.c.
3. If multiple 'c' or 'h' files or other libraries are used, place these files inside the folder.
4. The folder should have a) Word document (see template), b) source code file(s) with other 'c' and 'h' include files, c) text file with youtube video links (see template).
5. Submit the pdf file in canvas before the due date. The root folder of the github assignment directory should have the documentation and the text file with youtube video links.
6. Organize your youtube videos as playlist under the name "EMBSEC&ML". The playlist should have the video sequence arranged as submission or due dates.
7. Only submit pdf documents. Do not forget to upload this document in the github repository and in the canvas submission portal.

1. Goal: Explain what is explored in this assignment and what was accomplished?

In this assignment, the integration of X-NUCLEO-IKS4A1 with STM32F401RE-NUCLEO Board was explored. It involved working with LSM6DSV16X and LIS2MDL sensors to obtain accelerometer, gyroscope and magnetometer reading through UART. STM32Cube IDE was explored as the integrated development environment for coding the STM32F401 microcontroller. After completing this assignment, following things were accomplished:

- Learnt about hardware integration of X-NUCLEO-IKS4A1 with STM32F401RE-NUCLEO Board
 - Explored about STM32Cube IDE and how it is used for coding the STM32 microcontroller
 - Understood the way of debugging the codes in the microcontroller
 - Developed an understanding of the process to interact with the on-board sensors of NUCLEO-IKS4A1 and obtain the desired sensor readings in desired format
 - Learnt about the way to change different sensor parameters by modifying the sensor's header files
2. Screenshots of the IDE, physical setup, debugging process – Provide screenshot of successful compilation, screenshots of graphs, etc.

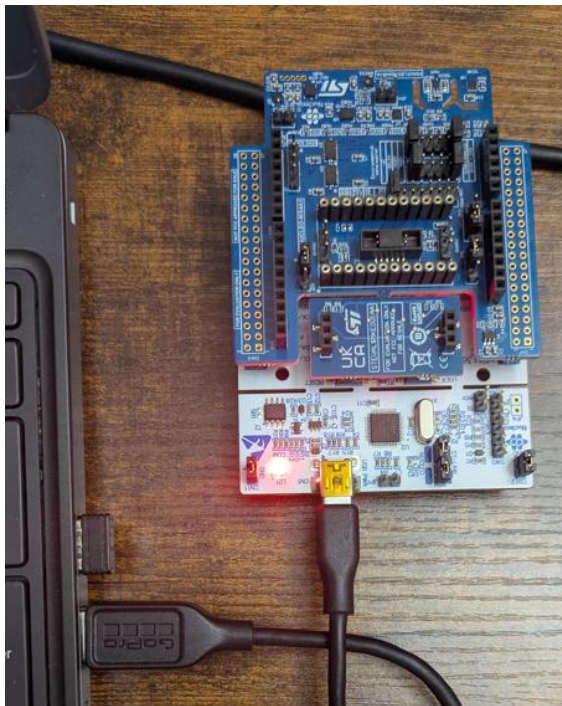


Figure 1: Physical setup of the STM32F401RE and X-NUCLEO-IKS4A1 board

Code Modifications

1. Changes to app_mems.c file

1.1 MX IKS4A1 DataLogTerminal Init() function

The yellow highlighted portion of the code represents the changes made to the code. All unnecessary sensor initialization and printing commands are commented out while keeping just the required sensor initialization.

[illegible]

```

        (int)out_value_odr.out_dec, (int)MotionCapabilities[i].AccMaxFS);
    printf("%s", dataOut);
    floatToInt(MotionCapabilities[i].GyroMaxOdr, &out_value_odr, 3);
    snprintf(dataOut, MAX_BUF_SIZE, " MAX GYRO ODR: %d.%03d Hz, MAX GYRO FS: %d\r\n",
(int)out_value_odr.out_int,
        (int)out_value_odr.out_dec, (int)MotionCapabilities[i].GyroMaxFS);
    printf("%s", dataOut);
    floatToInt(MotionCapabilities[i].MagMaxOdr, &out_value_odr, 3);
    snprintf(dataOut, MAX_BUF_SIZE, " MAX MAG ODR: %d.%03d Hz, MAX MAG FS: %d\r\n",
(int)out_value_odr.out_int,
        (int)out_value_odr.out_dec, (int)MotionCapabilities[i].MagMaxFS);
    printf("%s", dataOut);
    */
}
/*
IKS4A1_ENV_SENSOR_Init(IKS4A1_SHT40AD1B_0, ENV_TEMPERATURE | ENV_HUMIDITY);

IKS4A1_ENV_SENSOR_Init(IKS4A1_LPS22DF_0, ENV_TEMPERATURE | ENV_PRESSURE);

IKS4A1_ENV_SENSOR_Init(IKS4A1_STTS22H_0, ENV_TEMPERATURE);

for(i = 0; i < IKS4A1_ENV_INSTANCES_NBR; i++)
{
    IKS4A1_ENV_SENSOR_GetCapabilities(i, &EnvCapabilities[i]);
    snprintf(dataOut, MAX_BUF_SIZE,
        "\r\nEnvironmental Sensor Instance %d capabilities: \r\n TEMPERATURE: %d\r\n
PRESSURE: %d\r\n HUMIDITY: %d\r\n LOW POWER: %d\r\n",
        i, EnvCapabilities[i].Temperature, EnvCapabilities[i].Pressure,
EnvCapabilities[i].Humidity, EnvCapabilities[i].LowPower);
    printf("%s", dataOut);
    floatToInt(EnvCapabilities[i].TempMaxOdr, &out_value_odr, 3);
    snprintf(dataOut, MAX_BUF_SIZE, " MAX TEMP ODR: %d.%03d Hz\r\n",
(int)out_value_odr.out_int,
        (int)out_value_odr.out_dec);
    printf("%s", dataOut);
    floatToInt(EnvCapabilities[i].PressMaxOdr, &out_value_odr, 3);
    snprintf(dataOut, MAX_BUF_SIZE, " MAX PRESS ODR: %d.%03d Hz\r\n",
(int)out_value_odr.out_int,
        (int)out_value_odr.out_dec);
    printf("%s", dataOut);
    floatToInt(EnvCapabilities[i].HumMaxOdr, &out_value_odr, 3);
    snprintf(dataOut, MAX_BUF_SIZE, " MAX HUM ODR: %d.%03d Hz\r\n",
(int)out_value_odr.out_int, (int)out_value_odr.out_dec);
    printf("%s", dataOut);
}
*/
}

```

1.2 MX_IKS4A1_DataLogTerminal_Process() function

Here, unnecessary printing command and the portion with environmental sensors instances were commented out as they weren't required for this assignment.

```

void MX_IKS4A1_DataLogTerminal_Process(void)
{
    int32_t i;

```

[illegible]

1.3 Accelero Sensor Handler / Gyro Sensor Handler / Magneto Sensor Handler

This function (below) handles the printing of accelerometer sensor reading. Since we required the data to be printed in csv format, all the unnecessary printing lines are commented out as seen in the yellow highlight. Additionally, printing statement in our desired csv format was added as seen in the green highlight.

Same modification is done with Gyro_sensor_Handler and Magneto_Sensor Handler function.

```
static void Accelero_Sensor_Handler(uint32_t Instance)
{
    float odr;
    int32_t fullScale;
    IKS4A1_MOTION_SENSOR_Axes_t acceleration;
    displayFloatToInt_t out_value;
    uint8_t whoami;

    //snprintf(dataOut, MAX_BUF_SIZE, "\r\nMotion sensor instance %d:", (int)Instance);
    //printf("%s", dataOut);

    if (IKS4A1_MOTION_SENSOR_GetAxes(Instance, MOTION_ACCELERO, &acceleration))
    {
        snprintf(dataOut, MAX_BUF_SIZE, "\r\nACC[%d]: Error\r\n", (int)Instance);
    }
    else
    {
        //snprintf(dataOut, MAX_BUF_SIZE, "\r\nACC_X[%d]: %d, ACC_Y[%d]: %d, ACC_Z[%d]: %d\r\n",
        (int)Instance, (int)acceleration.x, (int)Instance, (int)acceleration.y, (int)Instance,
        (int)acceleration.z);
        snprintf(dataOut, MAX_BUF_SIZE, "\r\n%d, %d, %d, ", (int)acceleration.x,
        (int)acceleration.y, (int)acceleration.z);
    }

    printf("%s", dataOut);

    if (verbose == 1)
    {
        if (IKS4A1_MOTION_SENSOR_ReadID(Instance, &whoami))
        {
            snprintf(dataOut, MAX_BUF_SIZE, "WHOAMI[%d]: Error\r\n", (int)Instance);
        }
        else
        {
            snprintf(dataOut, MAX_BUF_SIZE, "WHOAMI[%d]: 0x%x\r\n", (int)Instance, (int)whoami);
        }

        printf("%s", dataOut);

        if (IKS4A1_MOTION_SENSOR_GetOutputDataRate(Instance, MOTION_ACCELERO, &odr))
        {
            snprintf(dataOut, MAX_BUF_SIZE, "ODR[%d]: ERROR\r\n", (int)Instance);
        }
        else
        {
            floatToInt(odr, &out_value, 3);
            snprintf(dataOut, MAX_BUF_SIZE, "ODR[%d]: %d.%03d Hz\r\n", (int)Instance,
            (int)out_value.out_int,
```

```

        (int)out_value.out_dec);
    }

    printf("%s", dataOut);

    if (IKS4A1_MOTION_SENSOR_GetFullScale(Instance, MOTION_ACCELERO, &fullScale))
    {
        snprintf(dataOut, MAX_BUF_SIZE, "FS[%d]: ERROR\r\n", (int)Instance);
    }
    else
    {
        snprintf(dataOut, MAX_BUF_SIZE, "FS[%d]: %d g\r\n", (int)Instance, (int)fullScale);
    }

    printf("%s", dataOut);
}
}

```

1.4 Temperature/Pressure/Humidity Sensor handlers

Full portion of code for temperature, pressure and humidity sensors' handlers were commented out as these aren't required for the current assignment.

2. Changes to lsm6dsv16x.c file

The LSM6DSV16X_Init() function inside lsm6dsv16x.c file contains the initialization parameter for the lsm6dsv16x sensor. Here, the full scale value of accelerometer is changed from 2g to 8g by changing LSM6DSV16X_2g to LSM6DSV16X_8g in the green highlight as shown. The fullscale for gyroscope was 2000 dbs by default and hence, it didn't need to be changed for the assignment requirement.

```

int32_t LSM6DSV16X_Init(LSM6DSV16X_Object_t *pObj)
{
    .....

    /* Full scale selection */
    if (lsm6dsv16x_xl_full_scale_set(&(pObj->Ctx), LSM6DSV16X_8g) != LSM6DSV16X_OK)
    {
        return LSM6DSV16X_ERROR;
    }

    .....
}

```

3. Changes to iks4a1_motion_sensors.h include file

3.1 Removal of 'if not defined' conditional statements for sensors

Here, except the sensors LSM6DSV16X and LIS2MDL, all other sensors were defined with value of zero since only those two sensors are required for the task. Furthermore, even though other

values were kept at zero, IKS4A1_MOTION_INSTANCES_NBR parameter (in app_mems.c; for loop) had the value of 4 for some reason. Hence, there is possibility that these values might be defined in other file already. Due to this, the 'if not defined' portion (ifndef) of the code was commented out for every sensors such that the values defined in this file are only considered. This kept the IKS4A1_MOTION_INSTANCES_NBR parameter value equal to 2 corresponding to our two sensors.

```
/* Includes -----*/
#include "iks4a1_conf.h"
#include "motion_sensor.h"

//ifndef USE_IKS4A1_MOTION_SENSOR_LSM6DSV16X_0
#define USE_IKS4A1_MOTION_SENSOR_LSM6DSV16X_0 1
//endif

//ifndef USE_IKS4A1_MOTION_SENSOR_LIS2DUXS12_0
#define USE_IKS4A1_MOTION_SENSOR_LIS2DUXS12_0 0
//endif

//ifndef USE_IKS4A1_MOTION_SENSOR_LIS2MDL_0
#define USE_IKS4A1_MOTION_SENSOR_LIS2MDL_0 1
//endif

//ifndef USE_IKS4A1_MOTION_SENSOR_LSM6DSO16IS_0
#define USE_IKS4A1_MOTION_SENSOR_LSM6DSO16IS_0 0
//endif

//ifndef USE_IKS4A1_MOTION_SENSOR_LSM6DSO_0
#define USE_IKS4A1_MOTION_SENSOR_LSM6DSO_0 0
//endif
```

3.2 Setting sensor instances

The sensor instances/indexes are defined inside **iks4a1_motion_sensors.h** header file as shown below. The LIS2MDL sensor instance was defined ahead of the LSM6DSV16X sensor due to which the magnetometer sensor got '0' index and the accelerometer/gyroscope sensor got the index of '1'. Hence, whenever the *MotionCapabilities[i]* array was encountered inside the for loop of **MX_IKS4A1_DataLogTerminal_Init()** and **MX_IKS4A1_DataLogTerminal_Process()** functions of app_mems.c file, it processed and printed the reading of magnetometer sensor first. However, as we require the sensor reading in the format of ax, ay, az, gx, gy, gz, mx, my, mz, the default index definition of both sensors were commented out (yellow highlight) and new sensor instance definition was added (green highlight) which simply defined the instance of LSM6DSV16X sensor as '0' and that of LIS2MDL sensor as '1'. This helped in printing the sensor reading in correct order.

```
#if (USE_IKS4A1_MOTION_SENSOR_LSM6DSO_0 == 1)
#define IKS4A1_LSM6DSO_0 0
#endif
```

```

#if (USE_IKS4A1_MOTION_SENSOR_LIS2DW12_0 == 1)
#define IKS4A1_LIS2DW12_0 (USE_IKS4A1_MOTION_SENSOR_LSM6DSO_0)
#endif

/*
#if (USE_IKS4A1_MOTION_SENSOR_LIS2MDL_0 == 1)
#define IKS4A1_LIS2MDL_0 (USE_IKS4A1_MOTION_SENSOR_LSM6DSO_0 + \
    USE_IKS4A1_MOTION_SENSOR_LIS2DW12_0)
#endif
*/
#if (USE_IKS4A1_MOTION_SENSOR_LIS2MDL_0 == 1)
#define IKS4A1_LIS2MDL_0 1
#endif

#if (USE_IKS4A1_MOTION_SENSOR_ASM330LHH_0 == 1)
#define IKS4A1_ASM330LHH_0 (USE_IKS4A1_MOTION_SENSOR_LSM6DSO_0 + \
    USE_IKS4A1_MOTION_SENSOR_LIS2DW12_0 + \
    USE_IKS4A1_MOTION_SENSOR_LIS2MDL_0)

#endif

.
.
.

/*
#if (USE_IKS4A1_MOTION_SENSOR_LSM6DSV16X_0 == 1)
#define IKS4A1_LSM6DSV16X_0 (USE_IKS4A1_MOTION_SENSOR_LSM6DSO_0 + \
    USE_IKS4A1_MOTION_SENSOR_LIS2DW12_0 + \
    USE_IKS4A1_MOTION_SENSOR_LIS2MDL_0 + \
    USE_IKS4A1_MOTION_SENSOR_ASM330LHH_0 + \
    USE_IKS4A1_MOTION_SENSOR_IIS2DLPC_0 + \
    USE_IKS4A1_MOTION_SENSOR_IIS2MDC_0 + \
    USE_IKS4A1_MOTION_SENSOR_ISM303DAC_ACC_0 + \
    USE_IKS4A1_MOTION_SENSOR_ISM303DAC_MAG_0 + \
    USE_IKS4A1_MOTION_SENSOR_ISM330DLC_0 + \
    USE_IKS4A1_MOTION_SENSOR_LIS2DH12_0 + \
    USE_IKS4A1_MOTION_SENSOR_LSM6DSOX_0 + \
    USE_IKS4A1_MOTION_SENSOR_AIS2DW12_0 + \
    USE_IKS4A1_MOTION_SENSOR_LIS3MDL_0 + \
    USE_IKS4A1_MOTION_SENSOR_LSM6DSR_0 + \
    USE_IKS4A1_MOTION_SENSOR_A3G4250D_0 + \
    USE_IKS4A1_MOTION_SENSOR_AIS328DQ_0 + \
    USE_IKS4A1_MOTION_SENSOR_AIS3624DQ_0 + \
    USE_IKS4A1_MOTION_SENSOR_H3LIS331DL_0 + \
    USE_IKS4A1_MOTION_SENSOR_LSM6DSRX_0 + \
    USE_IKS4A1_MOTION_SENSOR_ISM330DHCX_0 + \
    USE_IKS4A1_MOTION_SENSOR_LSM6DSO32_0 + \
    USE_IKS4A1_MOTION_SENSOR_IIS2ICLX_0 + \
    USE_IKS4A1_MOTION_SENSOR_AIS2IH_0 + \
    USE_IKS4A1_MOTION_SENSOR_LSM6DSO32X_0 + \
    USE_IKS4A1_MOTION_SENSOR_LSM6DSOX_SENSORHUB_LIS2MDL_0 + \
    USE_IKS4A1_MOTION_SENSOR_LIS2DU12_0 + \
    USE_IKS4A1_MOTION_SENSOR_ASM330LHHX_0)
#endif
*/

#if (USE_IKS4A1_MOTION_SENSOR_LSM6DSV16X_0 == 1)
#define IKS4A1_LSM6DSV16X_0 0
#endif

```

Successful Compilation

```
CDT Build Console [IKS4A1_DataLogTerminal]
02:18:18 **** Incremental Build of configuration Debug for project IKS4A1_DataLogTerminal ****
make -j16 all
arm-none-eabi-size IKS4A1_DataLogTerminal.elf
   text    data    bss     dec     hex filename
  42200     292    2548    45040    aff0 IKS4A1_DataLogTerminal.elf
Finished building: default.size.stdout

02:18:18 Build Finished. 0 errors, 0 warnings. (took 192ms)
```

Figure 2: Screenshot of successful build

```
Waiting for debugger connection...
Debugger connected
Waiting for debugger connection...
Debugger connected
Waiting for debugger connection...
-----
                STM32CubeProgrammer v2.20.0
-----

Log output file: C:\Users\dahalc1\AppData\Local\Temp\STM32CubeProgrammer_a23660.log
ST-LINK SN : 0671FF555185754867215634
ST-LINK FW : V2J46M33
Board : NUCLEO-F401RE
Voltage : 3.23V
SWD freq : 4000 KHz
Connect mode: Under Reset
Reset mode : Hardware reset
Device ID : 0x433
Revision ID : Rev Z
Device name : STM32F401xD/E
Flash size : 512 KBytes
Device type : MCU
Device CPU : Cortex-M4
BL Version : 0xD1

Opening and parsing file: ST-LINK_GDB_server_a23660.srec

Memory Programming ...
File : ST-LINK_GDB_server_a23660.srec
Size : 41.51 KB
Address : 0x08000000

Erasing memory corresponding to segment 0:
Erasing internal memory sectors [0 2]
Download in Progress:

File download complete
Time elapsed during download operation: 00:00:01.260

Verifying...

Time elapsed during verifying operation: 00:00:00.286

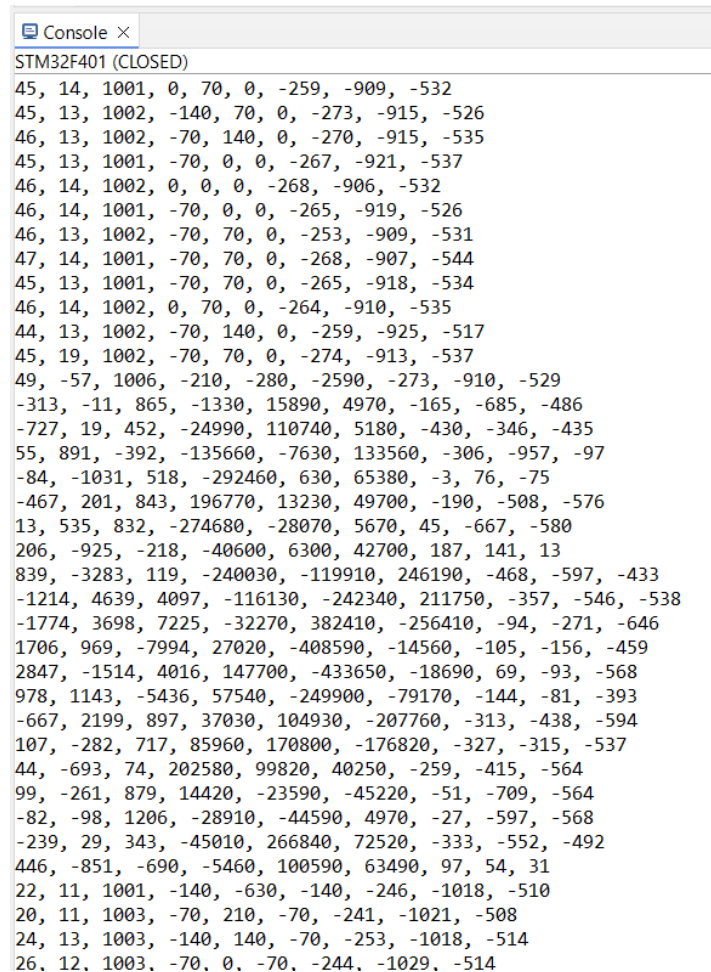
Download verified successfully

Shutting down...
Exit.
```

Figure 3: Screenshot of successful run

Output sensor reading

Figure 4 shows the sensor reading displayed in ax, ay, az, gx, gy, gz, mx, my, mz format. Initially, the sensors are at rest on a table and hence the sensor provide almost stable low values. After a while, the microcontroller board was shaken at high frequency and the change in reading can be seen (in the middle portion of output). As seen in the figure, the az value is fluctuating between +/- 8g since the full-scale parameter of accelerometer was changed to 8g. At the end, the board was again placed on the table and the sensor value again returned to the normal level.



```
STM32F401 (CLOSED)
45, 14, 1001, 0, 70, 0, -259, -909, -532
45, 13, 1002, -140, 70, 0, -273, -915, -526
46, 13, 1002, -70, 140, 0, -270, -915, -535
45, 13, 1001, -70, 0, 0, -267, -921, -537
46, 14, 1002, 0, 0, 0, -268, -906, -532
46, 14, 1001, -70, 0, 0, -265, -919, -526
46, 13, 1002, -70, 70, 0, -253, -909, -531
47, 14, 1001, -70, 70, 0, -268, -907, -544
45, 13, 1001, -70, 70, 0, -265, -918, -534
46, 14, 1002, 0, 70, 0, -264, -910, -535
44, 13, 1002, -70, 140, 0, -259, -925, -517
45, 19, 1002, -70, 70, 0, -274, -913, -537
49, -57, 1006, -210, -280, -2590, -273, -910, -529
-313, -11, 865, -1330, 15890, 4970, -165, -685, -486
-727, 19, 452, -24990, 110740, 5180, -430, -346, -435
55, 891, -392, -135660, -7630, 133560, -306, -957, -97
-84, -1031, 518, -292460, 630, 65380, -3, 76, -75
-467, 201, 843, 196770, 13230, 49700, -190, -508, -576
13, 535, 832, -274680, -28070, 5670, 45, -667, -580
206, -925, -218, -40600, 6300, 42700, 187, 141, 13
839, -3283, 119, -240030, -119910, 246190, -468, -597, -433
-1214, 4639, 4097, -116130, -242340, 211750, -357, -546, -538
-1774, 3698, 7225, -32270, 382410, -256410, -94, -271, -646
1706, 969, -7994, 27020, -408590, -14560, -105, -156, -459
2847, -1514, 4016, 147700, -433650, -18690, 69, -93, -568
978, 1143, -5436, 57540, -249900, -79170, -144, -81, -393
-667, 2199, 897, 37030, 104930, -207760, -313, -438, -594
107, -282, 717, 85960, 170800, -176820, -327, -315, -537
44, -693, 74, 202580, 99820, 40250, -259, -415, -564
99, -261, 879, 14420, -23590, -45220, -51, -709, -564
-82, -98, 1206, -28910, -44590, 4970, -27, -597, -568
-239, 29, 343, -45010, 266840, 72520, -333, -552, -492
446, -851, -690, -5460, 100590, 63490, 97, 54, 31
22, 11, 1001, -140, -630, -140, -246, -1018, -510
20, 11, 1003, -70, 210, -70, -241, -1021, -508
24, 13, 1003, -140, 140, -70, -253, -1018, -514
26, 12, 1003, -70, 0, -70, -244, -1029, -514
```

Figure 4: Sensor reading displayed in console

3. Declaration

I understand the Student Academic Misconduct Policy -
<http://studentconduct.unlv.edu/misconduct/policy.html>

"This assignment submission is my own, original work".
Chetan Dahal